Locally Honest Coverage Forces Nonparametric Inflation: a Sharpness Lower Bound for Prediction Sets

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Abstract

Marginal conformal coverage is compatible with oracle-rate sharpness and does not impose the bound studied here. A guarantee at a fixed test point is different. We show that any honestly valid procedure must, at a fixed smooth baseline, exceed the oracle threshold by the local modulus $(ng(x_0))^{-s/(2s+d)}$ on a majority of samples. The cause is Le Cam indistinguishability, a testing-power floor, not exchangeability. For fixed $\eta > 0$, minimax-rate sharpness stays compatible with honesty. Zero excess and adaptation to unknown smoothness are not. A plug-in estimate is sharper because it declines to insure against alternatives the data cannot rule out.

1 Two guarantees, one honest

Fix a model, freeze it, and let $R = A(X, Y)$ be its scalar nonconformity score with conditional law $\mathcal{L}(R \mid X = x)$. Ordinary conformal coverage is

$$\mathbb{P}\big\{R_{n+1} \in C_n(X_{n+1})\big\} \;\geq\; 1 - \alpha, \tag{1}$$

the probability averaging over the test point X_{n+1} and the calibration sample D_n. The honesty lower bound below does not follow from marginal validity: Lei and Wasserman (2014) construct bands with exact finite-sample marginal validity that converge to an oracle band at the minimax rate, and localized conformal methods retain marginal validity while weighting by proximity. The cost we identify attaches not to marginal coverage but to a guarantee at a fixed point, formalized next as honest local coverage.

Definition 1 (Honest local coverage)**.** Fix an interior point x_0 and $0 \leq \eta < \frac{1}{2}$. A one-sided prediction set $C_n(x_0) = [0, T_n(x_0)]$ for a nonnegative score is *honestly locally valid* over a class Θ if

$$\inf_{\theta \in \Theta} \mathbb{P}_\theta\Big\{F_{\theta(x_0)}\big(T_n(x_0)\big) \geq 1 - \alpha\Big\} \;\geq\; 1 - \eta, \tag{2}$$

where $F_{\theta(x_0)}$, assumed continuous and strictly increasing, is the true conditional law of the score at x_0, so the event equals $\{T_n(x_0) \geq q_\alpha(\theta(x_0))\}$, and the probability is over the calibration sample. In words: with probability at least $1 - \eta$ over the data, the realized set has conditional coverage $1 - \alpha$ at x_0, whatever law in Θ generated the data. Taking $\eta = 0$ is literal per-sample conditional coverage. This is a training-conditional, fixed-x_0 requirement, the local analogue of the training-conditional and tolerance-region guarantees studied for conformal sets (Vovk, 2012), and it is deliberately stronger than the marginal coverage (1).[1]

Sharpness is measured against the oracle threshold $q_\alpha(u) = F_u^{-1}(1-\alpha)$ through the excess

$$S_\theta(T_n) \;=\; \mathbb{E}_\theta\Big[\big(T_n(x_0) - q_\alpha(\theta(x_0))\big)_+\Big], \tag{3}$$

the expected inflation of the threshold above the oracle. The question is how small (3) can be made subject to (2).

2 Assumptions

The upper Hölder bound alone is not enough for a lower bound. The class must contain alternatives that are indistinguishable near x_0. Three standing assumptions supply this without asking us to estimate an arbitrary law.

Assumption 1 (Local design regularity)**.** The covariate density g satisfies $0 < g_- \le g(x) \le g_+ < \infty$ on a fixed neighbourhood of $x_0 \in (0,1)^d$, and x_0 is a Lebesgue point of g.

All the proof uses is the small-ball relation $\mathbb{P}_X\{X \in B(x_0,h)\} = \lambda(x_0)\,h^d(1+o(1))$, of which the density condition is the Euclidean special case with $\lambda(x_0) = g(x_0)$. The small-ball form also covers smooth manifolds, non-Euclidean metrics, and an intrinsic dimension d below the raw feature dimension. Without the Lebesgue point the local mass is only pinned between g_-h^d and g_+h^d, and every rate below holds with $g(x_0)$ replaced by a constant in $[g_-, g_+]$. The assumption fails at boundaries and in sparse high-dimensional regions, where the local sample size is smaller still and the inflation only grows.

Assumption 2 (Regular one-dimensional residual submodel)**.** The calibration sample $(X_i, R_i)_{i=1}^n$ is i.i.d. with $X_i \sim g$, and the admissible class of conditional score laws contains a one-parameter submodel $R \,|\, X = x \sim F_{\theta(x)}$, $\{F_u : u \in [\underline{u}, \overline{u}]\}$, that is, locally in u, quadratically regular in information and Lipschitz-regular in quantile:

$$D_{\mathrm{KL}}\big(F_u \,\|\, F_v\big) \le K\,(u-v)^2, \qquad c_q\,|u-v| \le \big|q_\alpha(u) - q_\alpha(v)\big| \le C_q\,|u-v|.$$

The lower bound needs only the quadratic information bound and the lower quantile constant c_q. The upper constant C_q enters only attainability (Section 4).

The submodel framing is the point. Even if the true residual law varies in scale, skewness, and tail thickness, the impossibility follows as soon as the class contains *one* regular direction that moves the target quantile, and a lower bound proved on the submodel applies to every larger class containing it. A sufficient condition is differentiability in quadratic mean with bounded Fisher information and $q_\alpha'(u)$ bounded above and away from zero, whence the mean-value theorem gives the quantile inequalities. The two-sided quantile bound forces q_α strictly monotone, so we orient the parameter to make it increasing. What is excluded is worth naming:

[1] *Honest* is the standard term for a coverage guarantee that holds uniformly over the class at finite n, the $\inf_\theta$ sitting inside the probability, as opposed to a pointwise or asymptotic guarantee that fixes θ first (Li, 1989; Low, 1997; Cai and Low, 2004). Here *local* means the requirement is imposed at the fixed point x_0 rather than over an interval.

discrete scores, quantiles in flat parts of a CDF, parameters with $q'_\alpha(u) = 0$, and families with parameter-dependent support giving infinite D_{KL}.

Three standard scores satisfy the submodel exactly on a compact parameter interval $[\underline{u}, \overline{u}]$, with $z_p = \Phi^{-1}(p)$ and $c_\alpha = F^{-1}_{\chi^2_1}(1-\alpha)$.

(A) *Heteroskedastic squared-Gaussian score.* $E \mid X = x \sim N(0, e^{2u})$, $R = E^2 \sim e^{2u}\chi^2_1$. Then $q_\alpha(u) = c_\alpha e^{2u}$ and $D_{\mathrm{KL}}(F_u \| F_v) = \frac{1}{2}[e^{2(u-v)} - 1 - 2(u-v)] \le e^{2\rho}(u-v)^2$ for $|u-v| \le \rho$. The result therefore already applies to ordinary heteroskedastic Gaussian errors and squared residuals.

(B) *Lognormal positive score.* $R = \exp\{u + \sigma Z\}$, $Z \sim N(0,1)$. Then $D_{\mathrm{KL}}(F_u \| F_v) = (u - v)^2/(2\sigma^2)$ exactly and $q_\alpha(u) = \exp\{u + \sigma z_{1-\alpha}\}$, so both bounds hold with $c_q = e^{\underline{u}+\sigma z_{1-\alpha}}$, $C_q = e^{\overline{u}+\sigma z_{1-\alpha}}$. This model gives a rigorous attainability construction (Section 4) and the exact numerical check (Section 7).

(C) *Exponential scale.* $R \mid u \sim \mathrm{Exp}(\text{mean } e^u)$. Then $D_{\mathrm{KL}}(F_u \| F_v) = e^{u-v} - 1 - (u-v)$ and $q_\alpha(u) = -(\log \alpha)e^u$; both bounds follow on a compact interval, with no truncation needed.

Assumption 3 (Hölder smoothness with local richness)**.** Write $s = k + \gamma$ with $k = \lfloor s \rfloor$ and $0 < \gamma \le 1$, and let $\mathcal{H}^s(L)$ be the ball of fields θ with kth derivatives γ-Hölder with seminorm $\le L$ and range $\theta(x) \in [\underline{u}, \overline{u}]$. Fix a bump profile ψ with $\psi(0) = 1$, $\|\psi\|_\infty \le 1$, $\operatorname{supp}\psi \subseteq B(0,1)$, and $\psi \in C^s$. There are $h_0, a_0 > 0$ and an interior baseline value $u_0 \in (\underline{u}, \overline{u})$ with

$$\theta_{h,a}(x) = u_0 + a\,L\,h^s\,\psi\Big(\frac{x - x_0}{h}\Big) \in \mathcal{H}^s(L), \qquad 0 < h < h_0, \quad 0 < a < a_0.$$

The scaling $aLh^s\psi((x - x_0)/h)$ has C^s seminorm $\asymp aL$ independent of h, so small a keeps it in the ball.

Assumption 3 is the operative one, and it is not innocuous. It holds for broad Hölder classes that permit localized features at the detection boundary, but it can fail under global parametric structure, analyticity, monotonicity, or self-similarity, and those are exactly the restrictions that can restore honest adaptation. Within a Hölder ball it says the class holds a pair that agree outside a ball of radius h around x_0 but differ by $\delta = aLh^s$ at x_0, with the amplitude a shrinkable below a_0, which the proof uses to place the two hypotheses at any chosen separation.

3 The lower bound

The bound is at bottom a testing-power statement, so we isolate that step. For laws P_1, P_0 on the calibration sample write the Neyman–Pearson frontier

$$\beta_{1-\eta}(P_1, P_0) \;=\; \inf_{A:\, P_1(A) \ge 1-\eta} P_0(A), \tag{4}$$

the least P_0-mass of any event whose P_1-mass is at least $1 - \eta$.

Lemma 2 (Honesty is a test, and a test has a floor)**.** *Let θ_0, θ_1 be two parameter fields with $q_1 := q_\alpha(\theta_1(x_0)) > q_\alpha(\theta_0(x_0)) =: q_0$, and let T_n satisfy honest local coverage* (2) *under θ_1. Then*

$$\mathbb{P}_{\theta_0}\big\{T_n(x_0) \ge q_1\big\} \;\ge\; \beta_{1-\eta}\big(P^n_{\theta_1}, P^n_{\theta_0}\big), \qquad S_{\theta_0}(T_n) \;\ge\; (q_1 - q_0)\,\beta_{1-\eta}\big(P^n_{\theta_1}, P^n_{\theta_0}\big).$$

Proof. Honesty under θ_1 makes $A = \{T_n(x_0) \geq q_1\}$, a function of the calibration sample, satisfy $P^n_{\theta_1}(A) \geq 1-\eta$, so $P^n_{\theta_0}(A) \geq \beta_{1-\eta}$ by (4). On A the excess $T_n(x_0) - q_0$ is at least $q_1 - q_0$, giving the second inequality. The frontier obeys the Le Cam bound $\beta_{1-\eta}(P_1, P_0) \geq 1 - \eta - \mathrm{TV}(P_1, P_0)$, since any A with $P_1(A) \geq 1-\eta$ has $P_0(A) \geq P_1(A) - \mathrm{TV}$. □

The frontier (4) is exact; the $1 - \eta - \mathrm{TV}$ reading is the crude corollary. Both feed the same conclusion once the two hypotheses are placed at critical separation.

Theorem 3 (Honest local coverage forces inflation)**.** *Fix $0 < \alpha < 1$ and $0 \leq \eta < \frac{1}{2}$, and let Assumptions 1–3 hold. There are constants $c_1, c_2 > 0$ and n_0, depending only on $(\alpha, \eta, K, c_q, g_+, \psi, a_0, h_0)$ and the distance from u_0 to the parameter boundary, and a fixed constant baseline $\theta_0 \in \mathcal{H}^s(L)$, such that for all $n \geq n_0$ every T_n satisfying honest local coverage (2) over $\mathcal{H}^s(L)$ has, with $r_n := L^{d/(2s+d)}\big(n\, g(x_0)\big)^{-s/(2s+d)}$,*

$$\mathbb{P}_{\theta_0}\big\{\, T_n(x_0) - q_\alpha(\theta_0(x_0)) \;\geq\; c_1\, r_n \,\big\} \;\geq\; c_2 \;>\; \tfrac{1}{2}. \tag{5}$$

In particular the realized threshold exceeds the oracle by order r_n on a majority of samples, so both its median excess and its expected positive excess satisfy

$$\sup_{\theta \in \mathcal{H}^s(L)} S_\theta(T_n) \;\geq\; S_{\theta_0}(T_n) \;\geq\; c_1 c_2\, r_n. \tag{6}$$

Proof. Take the constant baseline $\theta_0 \equiv u_0$ and the bump $\theta_1 = \theta_{h,a}$ of Assumption 3. Since $\psi(0) = 1$ and q_α is oriented increasing (Assumption 2), $q_1 = q_\alpha(u_0 + \delta) > q_\alpha(u_0) = q_0$; if $\psi(0) = 1$ gave a decrease we would flip the sign of ψ. Both fields lie in $\mathcal{H}^s(L)$.

Indistinguishability. The two laws differ only on the bump support, of covariate measure $g(x_0)h^d(1+o(1))$ by Assumption 1. Integrating the per-point divergence bound $D_{\mathrm{KL}}(F_{\theta_1(x)} \| F_{u_0}) \leq K\delta^2\psi((x-x_0)/h)^2$ of Assumption 2 against the i.i.d. sample,

$$D_{\mathrm{KL}}\big(P^n_{\theta_1} \,\|\, P^n_{\theta_0}\big) \;\leq\; n\, K\, a^2 L^2 h^{2s+d}\, g(x_0) \int \psi(z)^2\, dz\, (1 + o(1)), \qquad \delta = a\, L\, h^s.$$

Fix the bandwidth by $h = c_h\big(L^2 n g(x_0)\big)^{-1/(2s+d)}$. The divergence bound becomes $K a^2 c_h^{2s+d} \|\psi\|_2^2 (1 + o(1))$, free of n, and the quantile shift is

$$q_1 - q_0 \;\geq\; c_q\, \delta \;=\; c_q\, a\, L\, h^s \;=\; c_q\, a\, c_h^s\, L^{d/(2s+d)}\big(n g(x_0)\big)^{-s/(2s+d)} \;=\; c_1\, r_n,$$

linear in a, using $1 - \frac{2s}{2s+d} = \frac{d}{2s+d}$ so $\delta \asymp r_n$. Now shrink $a < a_0$ (a constant, not depending on n) until Pinsker gives $\mathrm{TV}(P^n_{\theta_1}, P^n_{\theta_0}) \leq \tau$ with $\tau < \frac{1}{2} - \eta$, which is possible because the divergence bound scales as a^2. Then Lemma 2 gives $\beta_{1-\eta} \geq 1 - \eta - \tau =: c_2 > \frac{1}{2}$, and $c_1 = c_q a c_h^s$ depends on the chosen a.

Inflation. By the first part of Lemma 2, $\mathbb{P}_{\theta_0}\{T_n(x_0) \geq q_1\} \geq c_2$, and $\{T_n(x_0) \geq q_1\} \subseteq \{T_n(x_0) - q_0 \geq c_1 r_n\}$ because $q_1 - q_0 \geq c_1 r_n$, which is (5). Taking the expected positive part gives $S_{\theta_0}(T_n) \geq c_1 c_2\, r_n$, and $\theta_0 \in \mathcal{H}^s(L)$ puts the supremum in (6) at least this high. The threshold n_0 is where the bump support fits inside the neighbourhood of Assumptions 1–3 and $\theta_{h,a}$ stays in the parameter interval. □

The mechanism is the intended one. The data cannot tell a smooth baseline from a small, local increase in residual uncertainty. Exact local protection against the increase forces the set to stay inflated even when the baseline is true, and the size of the forced inflation is the amplitude of the largest undetectable bump. The statement (5) is stronger than an expectation bound: it is the *typical* realized threshold, not an average over rare wide sets, that carries the inflation.

4 Attainability in a regular example

An honest procedure attains the lower-bound rate, so Theorem 3 is a real optimum and not an empty constraint. We show this in the lognormal model (B), where the finite-sample honesty statement is exact rather than an O_P assertion. Assume $0 < s \le 1$, $0 < \eta < \frac{1}{2}$, and

$$\log R_i \;=\; \theta(X_i) + \varepsilon_i, \qquad \varepsilon_i \sim_{\text{iid}} N(0,1), \qquad \theta \in \mathcal{H}^s(L).$$

Let $N_k(x_0)$ be the k nearest calibration points, let ρ_k be the distance from x_0 to its kth nearest point, and let $\widehat{\theta}_k = k^{-1}\sum_{i\in N_k(x_0)} \log R_i$. Set

$$T_{n,k}(x_0) \;=\; \exp\Big\{\,\widehat{\theta}_k \;+\; L\rho_k^s \;+\; k^{-1/2} z_{1-\eta} \;+\; z_{1-\alpha}\,\Big\}. \tag{7}$$

Proposition 4 (Exact honest attainment)**.** *For every $\theta \in \mathcal{H}^s(L)$, the set $[0, T_{n,k}(x_0)]$ is honestly locally valid (2) conditional on the design, and with $k \asymp L^{-2d/(2s+d)}\big(ng(x_0)\big)^{2s/(2s+d)}$ its excess is $O(r_n)$.*

Proof. Conditional on $X_1,\dots,X_n$, $\widehat{\theta}_k = k^{-1}\sum_{i\in N_k}\theta(X_i) + N(0,1/k)$, and Hölder smoothness with $s \le 1$ gives $|\theta(x_0) - k^{-1}\sum_{i\in N_k}\theta(X_i)| \le L\rho_k^s$. The oracle threshold is $q_\alpha(\theta(x_0)) = \exp\{\theta(x_0) + z_{1-\alpha}\}$, so $T_{n,k}(x_0) \ge q_\alpha(\theta(x_0))$ iff the standard normal $Z = \sqrt{k}(\widehat{\theta}_k - k^{-1}\sum\theta(X_i))$ satisfies $Z \ge \sqrt{k}\,(\theta(x_0) - k^{-1}\sum\theta(X_i) - L\rho_k^s) - z_{1-\eta}$. The right side is $\le -z_{1-\eta}$, and $\mathbb{P}\{Z \ge -z_{1-\eta}\} = 1-\eta$, giving conditional coverage at least $1-\eta$ for every θ. Under Assumption 1, $\rho_k^s \asymp (k/ng(x_0))^{s/d}$, so the excess is of order $L(k/ng(x_0))^{s/d} + k^{-1/2}$, and the stated k balances the two terms at $L^{d/(2s+d)}(ng(x_0))^{-s/(2s+d)} = r_n$. □

Three points. The construction is a genuine finite-sample, design-conditional honesty statement, not a concentration theorem asserted without proof. It needs $\eta > 0$: the margin $k^{-1/2}z_{1-\eta}$ diverges as $\eta \to 0$. And it uses $s \le 1$, where a local average has bias $O(L\rho_k^s)$. For $s > 1$ the local average is replaced by a local-polynomial estimate of degree $\lfloor s \rfloor$, which restores the bias order.

Remark 5 (The endpoint $\eta = 0$)*.* The modulus rate is an $\eta > 0$ statement. At $\eta = 0$, exact per-sample conditional validity is far more expensive. If the finite-sample laws are mutually absolutely continuous, as in examples (A)–(C), honest validity transfers every probability-one upper-threshold event between parameter values: $P^n_{\theta_1}\{T_n(x_0) \ge q_1\} = 1$ forces $P^n_{\theta_0}\{T_n(x_0) \ge q_1\} = 1$. Taking a *fixed* nonvanishing bump from Assumption 3 then forces a constant, non-shrinking inflation at the baseline. So the honest optimum is the modulus r_n only for fixed $\eta > 0$. Exact $\eta = 0$ validity costs a constant, or a vacuous threshold.

5 The bandwidth reading

The rate is the local modulus of continuity,

$$r_n \;=\; L^{d/(2s+d)}\big(ng(x_0)\big)^{-s/(2s+d)} \;\asymp\; \inf_{h>0}\Big\{\,L\,h^s \;+\; \big(n\,g(x_0)\,h^d\big)^{-1/2}\,\Big\}, \tag{8}$$

the sum of the largest undetectable local heterogeneity, Lh^s, and the local sampling scale, $(ng(x_0)h^d)^{-1/2}$, minimized at $h \asymp (L^2 ng(x_0))^{-1/(2s+d)}$. Writing $m_h \asymp ng(x_0)h^d$ for the effective local sample size, the honesty cost is $Lh^s + m_h^{-1/2}$: as data thin, both the best neighbourhood and the unavoidable inflation grow. This is the small-sample statement, with no appeal to uniform pooling. Exact local validity does not force wide neighbourhoods; it forces protection against locally undetectable alternatives, whose width is the modulus.

6 Unknown smoothness: honesty cannot adapt

Over a single known class the lower bound (6) is attainable, so coverage and minimax-rate sharpness are not incompatible there. The genuine separation is adaptive.

Theorem 6 (No adaptation under honesty)**.** *Let $0 < s_0 < s_1$, fix $0 < \eta < \frac{1}{2}$, and require honest local coverage over $\mathcal{H}^{s_0}(L) \cup \mathcal{H}^{s_1}(L)$. Then there is $c > 0$ with*

$$\sup_{\theta \in \mathcal{H}^{s_1}(L)} S_\theta(T_n) \;\geq\; c\, L^{d/(2s_0+d)} \big(ng(x_0)\big)^{-s_0/(2s_0+d)}, \tag{9}$$

which is strictly slower than the smooth-class rate $L^{d/(2s_1+d)}\big(ng(x_0)\big)^{-s_1/(2s_1+d)}$. An honest set cannot achieve the faster sharpness available on the smoother subclass.

Proof. Take the baseline $\theta_0 \equiv u_0$, which is constant and hence lies in $\mathcal{H}^{s_1}(L)$, and the indistinguishable bump alternative from the rougher class $\mathcal{H}^{s_0}(L)$ at the s_0-critical bandwidth. The proof of Theorem 3 applies verbatim and forces $S_{\theta_0}(T_n) \gtrsim L^{d/(2s_0+d)}(ng(x_0))^{-s_0/(2s_0+d)}$, even though $\theta_0 \in \mathcal{H}^{s_1}(L)$. □

This is the honesty–adaptation obstruction familiar from nonparametric confidence bands (Low, 1997; Cai and Low, 2004). A sharp point estimator can read apparent smoothness off the data and choose a favourable bandwidth. An honest coverage guarantee cannot trust apparent smoothness, because a rough local alternative is statistically indistinguishable, so it must carry the rougher width. This is the strongest form of the thesis: the local estimator is sharper not by a better estimate but by declining to insure against what it cannot see.

Remark 7 (Average validity is weaker; a heuristic). If validity means only $\inf_\theta \mathbb{E}_\theta F_{\theta(x_0)}(T_n) \geq 1-\alpha$, averaging over calibration samples, a procedure may compensate a too-narrow set on some samples with a wide set on others, and the first-order argument weakens. The following is a sketch, not a proved bound: it needs the random threshold to stay, with high probability, in a region where $t \mapsto F_u(t)$ is uniformly strongly concave, which we do not verify here. Under that condition Jensen gives $\mathbb{E}_\theta T_n \geq q_\alpha(\theta(x_0))$ and a curvature bound $\mathrm{Var}(T_n) \lesssim \mathbb{E}_\theta T_n - q_\alpha(\theta(x_0))$, so small expected excess would make T_n a low-MSE estimator of the conditional quantile, and the minimax MSE bound then yields $\sup_\theta[\mathbb{E}_\theta T_n - q_\alpha(\theta(x_0))] \gtrsim (ng(x_0))^{-2s/(2s+d)}$. The exponent would double because average coverage is a materially weaker requirement than honest per-sample coverage.

7 An exact check

The bound is a testing statement, so in the lognormal model (B) it has no free constants to estimate. Take fixed design $x_i = (i - \frac{1}{2})/n$ and nonnegative scores $R_i = \exp\{\theta(x_i) + Z_i\}$, $Z_i \sim N(0,1)$, baseline $\theta_0 \equiv 0$, and the triangular bump $\psi(z) = (1-|z|)_+$ with $s = d = 1$. The map $R \mapsto \log R$ is a bijection, so the log-likelihood ratio, and hence every testing quantity, is exactly that of the Gaussian location model: with separation $I_n = \sum_i (\theta_1(x_i) - \theta_0(x_i))^2$,

$$\mathrm{TV}(P^n_{\theta_1}, P^n_{\theta_0}) = 2\Phi\big(\tfrac{1}{2}\sqrt{I_n}\big) - 1, \qquad \beta_{1-\eta}(P^n_{\theta_1}, P^n_{\theta_0}) = \Phi\big(z_{1-\eta} - \sqrt{I_n}\big),$$

the second being the exact frontier (4). The quantile shift is $q_1 - q_0 = e^{z_{1-\alpha}}(e^\delta - 1) \sim e^{z_{1-\alpha}}\delta$, so the critical exponent and the constant inflation fraction are unchanged. Along the critical

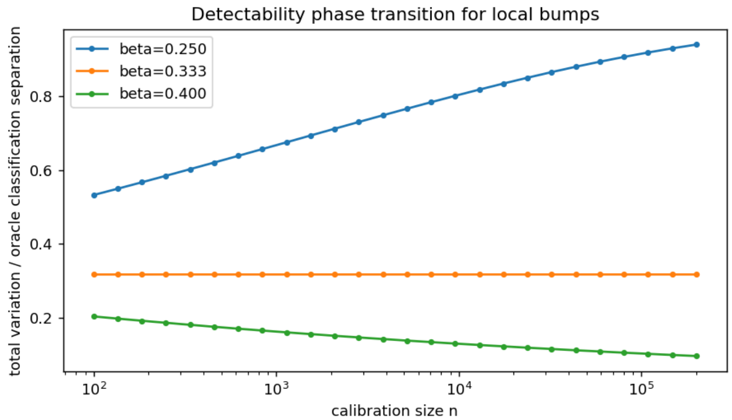

Figure 1: Total variation between the two n-sample laws, for a bump bandwidth $h = n^{-\beta}$. At the critical $\beta = 1/(2s + d) = 1/3$ it is flat at 0.317. Below it rises to one (detectable), above it falls to zero (invisible).

scaling $h = n^{-1/(2s+d)}$, $\delta = h^s$, the separation converges, $I_n \to \int_{-1}^{1}(1 - |z|)^2\, dz = \frac{2}{3}$, so both quantities are constant in n: the Le Cam boundary. Below the critical exponent the bump is detectable and TV $\to 1$; above it, invisible and TV $\to 0$ (Figure 1). At the boundary the forced minimum inflation, as a fraction of the quantile shift, is $\beta_{0.9} = \Phi(z_{0.9} - \sqrt{2/3}) = 0.679$, strictly above the crude Le Cam reading $1 - \eta - \mathrm{TV} = 0.583$ and, being above $\frac{1}{2}$, forcing inflation on a majority of samples (Figure 2). The nonadaptation gap of Theorem 6 shows the same way: the forced s_0-rate over the smooth-s_1 oracle rate diverges as $n^{s_1/(2s_1+d)-s_0/(2s_0+d)} = n^{1/15}$, matched to four decimal places by the closed form. Figure 3 runs the honest k-nearest-neighbour band (7) against the uncorrected plug-in threshold: the honest band holds $1 - \eta$ coverage at the baseline while the plug-in covers about half the time, at the cost of the r_n inflation the theorem predicts. A random-design exponential experiment reproduces every scaling by Monte Carlo.

8 What it says for conformal prediction

The obstruction is not exchangeability, weighting, or pooling. Localized conformal sets that reweight by proximity are marginally valid and can be sharp. The obstruction is indistinguishability under an honesty requirement. Three statements should be kept apart. Zero or $o(r_n)$ oracle excess is incompatible with honesty. Minimax-rate sharpness, $O(r_n)$ excess at a known smoothness s, is compatible with honesty *for fixed* $\eta > 0$, and Section 4 attains it; at $\eta = 0$ even that fails (Remark 5). The smooth-class rate while honest over a rougher class is again incompatible. Theorem 3 is therefore a nonparametric *inflation* theorem, not a claim that honesty and minimax-rate sharpness never coexist, and Theorem 6 is the genuine separation.

The applied reading is the one experience already supplies. At small effective local sample size m_h, the modulus $Lh^s + m_h^{-1/2}$ is large and the honest set is inflated by a correspondingly large amount. When the objective is a sharp predictive score rather than high-probability local coverage, a kernel or local-likelihood estimate of the residual law may be preferable, because it does not carry the one-sided insurance radius honesty requires. Whether it wins is a separate decision-theoretic comparison under the chosen score, not a corollary of the width bound. The

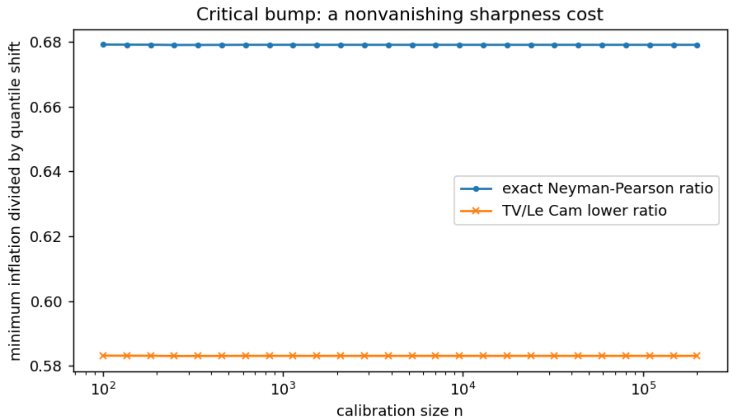

Figure 2: Minimum inflation forced at the smooth baseline, as a fraction of the quantile shift. The exact Neyman–Pearson frontier (0.679) sits well above the crude Le Cam/TV bound (0.583), and both are constant in n and above $\frac{1}{2}$.

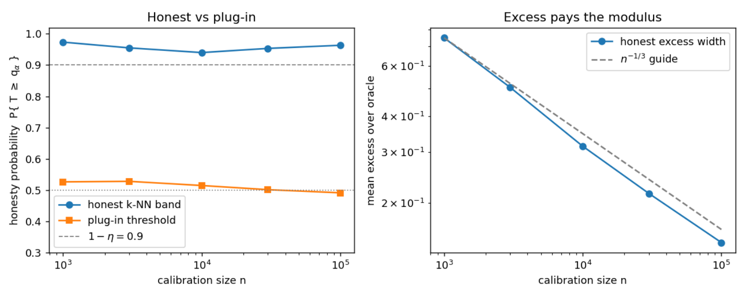

Figure 3: Honest k-nearest-neighbour band (7) versus the uncorrected plug-in threshold, lognormal scores at the smooth baseline. The honest band holds coverage at $1-\eta=0.9$; the plug-in covers about half the time. The gap is the r_n inflation the lower bound forces, and it decays at the predicted $n^{-1/3}$.

honest guarantee is not free and it is not distribution-free at a point. Its price is the sharpness it surrenders to insure against the alternatives the data cannot rule out.

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